Department of Information Engineering

DIPARTIMENTO DI INGEGNERIA DELL'INFORMAZIONE

LM-IDNet: Adaptive IoT Network Anomaly Detection

Dirichlet–multinomial anomaly detection

Languasco–Migliardi likelihood evaluation

Master's Degree in Computer Engineering

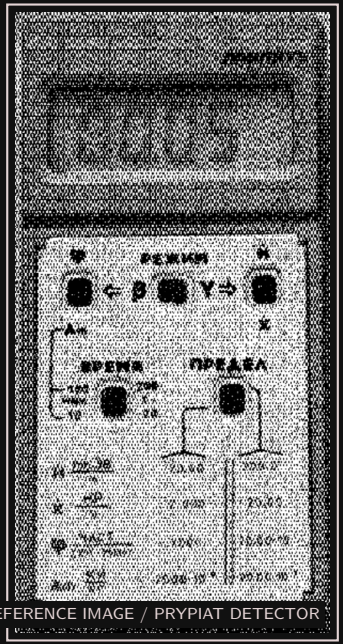
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Co-supervisor: Professor Alessandro Languasco

Scientific support: Professor Sherenaz Al-Haj Baddar

Academic Year 2025/2026



REFERENCE IMAGE / PRYPIAT DETECTOR

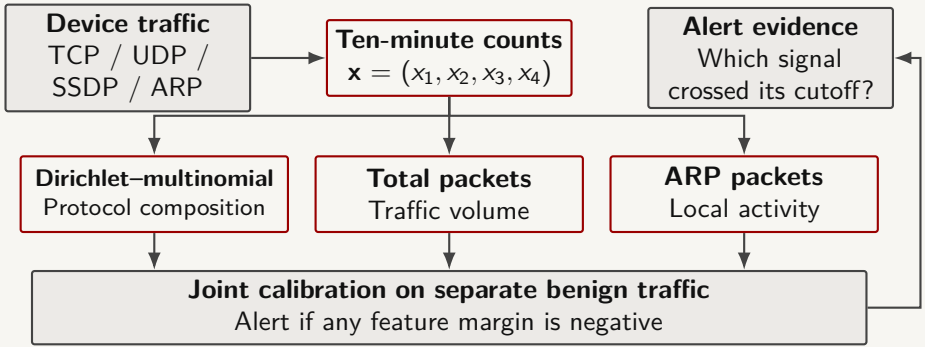
Context: IoT anomaly-detection approaches

Research studies	Operational platforms	Scope of this study
N-BaIoT, Kitsune and DÏoT use richer neural, ensemble or federated representations.	Agentless discovery, broad telemetry, threat intelligence and SOC response.	Four explicit counts, a device-specific normal model and inspectable decision margins.
Strong modelling capacity, but different data and protocols prevent a direct metric ranking.	Microsoft Defender for IoT and Palo Alto Device Security address a wider operational scope.	Dirichlet–multinomial scoring, numerical validation and a constrained-runtime evaluation.

LM-IDNet proposes device-specific detection from four counts using a Dirichlet–multinomial (DM) model.

It combines composition, traffic-volume and ARP checks with calibrated margins and guarded periodic refitting.

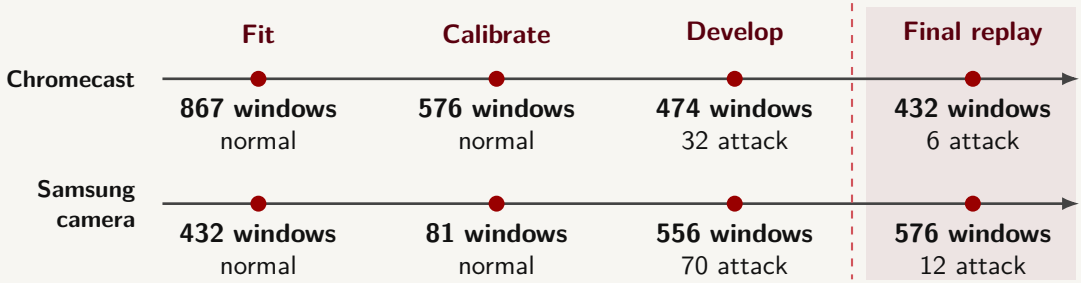
Four counts, three signals, one explainable decision



The Languasco–Migliardi (LM) algorithm evaluates the DM likelihood using stable log-gamma differences.

Concentration measures how tightly traffic stays around its average protocol mix; ψ is its inverse. A smaller ψ is closer to the multinomial limit, where direct log-gamma subtraction can lose precision.

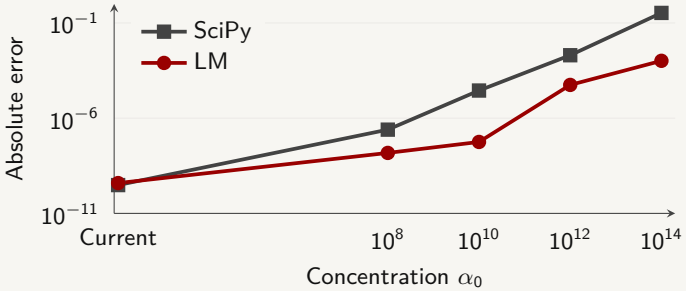
Datasets and chronological partitions



ICNC 2025 ALM baseline: $\psi < 0.05$; Chromecast passes, Samsung does not.

Reference: Al-Haj Baddar, Languasco and Migliardi. Values: 0.0356 / 0.0575; both show extra-multinomial variation. Labels are used only for metrics.

Numerical comparison: LM and SciPy



$\sim$500$\times$
lower absolute error at $\alpha_0 = 10^{10}$

$\sim$327$\times$
lower absolute error at $\alpha_0 = 10^{14}$

Current data: 0/1300 decision differences.

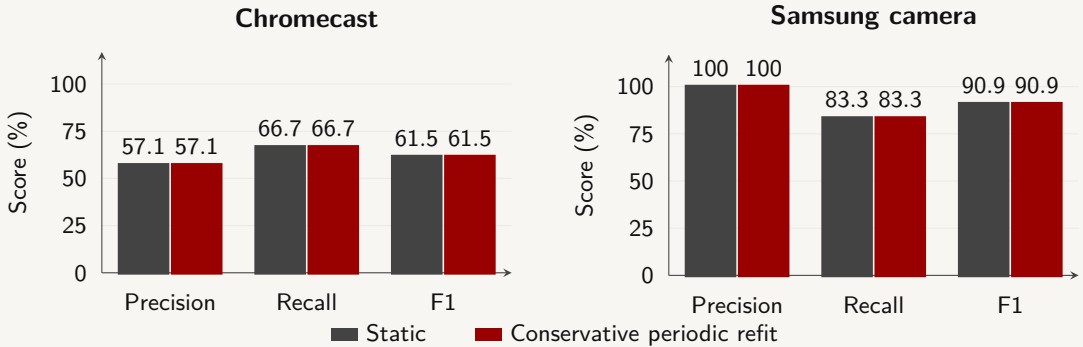
Live score: LM 12.04 μs vs SciPy 13.39 μs (10.1% faster). **Fit/refit:** LM 193 ms vs SciPy 181 ms (LM 6.4% slower).

Both agree on current data; LM has lower error in the high-concentration stress regime.

100-digit reference probe. Runtime is workload-specific; ten-minute collection dominates alert delay.

Final detection results by device

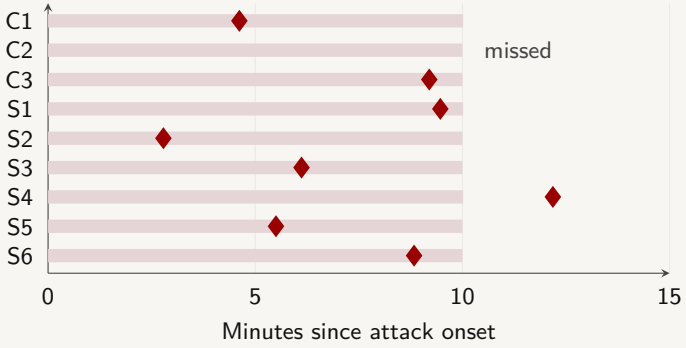
Ten-minute windows ▪ Final partitions



Chromecast: 4/6 attack windows flagged. Samsung camera: 10/12 flagged.

Periodic-refit results match static results for each device; no candidate update was activated.

Attack coverage and detection delay



First alert

Annotated attack

Chromecast

2/3 detected; 2/3 by end

6.9 min median

Samsung camera

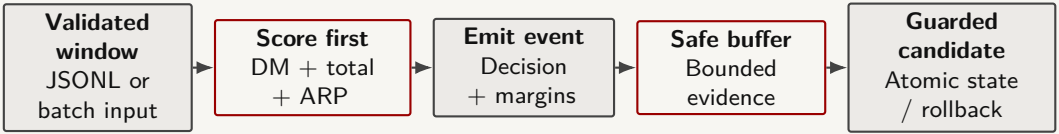
6/6 detected; 5/6 by end

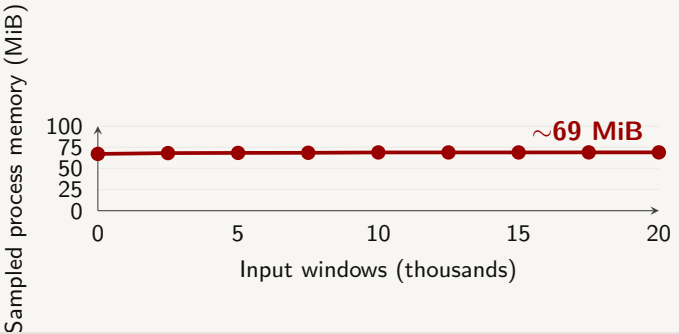
7.5 min median

The ten-minute collection interval is part of the detection delay.

Static mode, final official intervals; C = Chromecast, S = Samsung. Median delay is among detected attacks; alerts occur at window close.

Edge architecture and simulation





20,000

inputs: 16,000 scored; 4,000 missing skipped

~69 MiB

sampled process memory

5.079 s worst observed window, including a rejected refit

One detector serves batch and streaming; updates affect only later windows.

Accelerated synthetic replay; x86_64 Docker, 1 CPU / 512 MiB. No candidate update was promoted.

Summary: results and limitations

Observed results

- Four counts support three inspectable alert signals.
- 8/9 attack intervals were alerted; 7 by attack end.
- LM reduces absolute error in the high-concentration stress regime.
- Streaming remained near 69 MiB with guarded persistent state.

Scope and limitations

- Two labelled devices and 18 final attack windows.
- Count-preserving attacks may remain invisible.
- The edge trial is synthetic x86_64, not a physical gateway.
- Useful conservative adaptation remains unproven on these traces.

The prototype combines explainable detection, stable likelihood evaluation and guarded runtime state.

Claims remain limited to the recorded datasets, numerical probe and constrained simulation.



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Thank you

Questions and discussion

LM-IDNet ▪ Fatemeh Mahvari

Reference slides follow: model, profiles, native method, counts, guards, window study, sources.

R1 / Model and decision rule

DM composition model $\alpha_k > 0$, $\alpha_0 = \sum_k \alpha_k$, $N = \sum_k x_k$

$$\ell(\mathbf{x}) = \log \frac{N!}{\prod_k x_k!} + \log \frac{\Gamma(\alpha_0)}{\Gamma(\alpha_0 + N)} + \sum_k \log \frac{\Gamma(\alpha_k + x_k)}{\Gamma(\alpha_k)}.$$

Three signals $\ell(\mathbf{x})$, $-\log(1 + N)$, $-\log(1 + x_{\mathrm{ARP}})$.

Fit-only medians and IQRs scale each signal. The minimum scaled signal is calibrated jointly at the 1% lower quantile on separate benign windows.

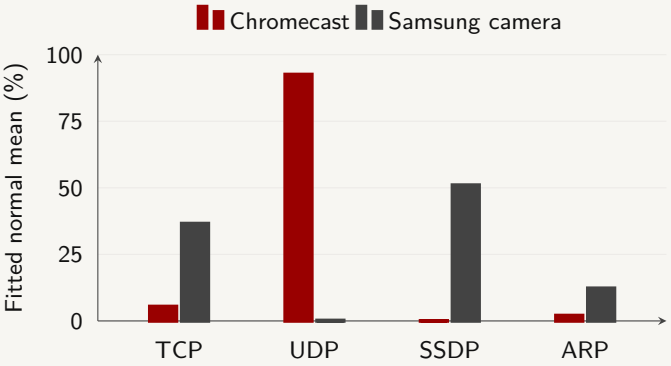
The event score is the minimum signed margin from the resulting cutoffs:

$$s = \min(m_{\mathrm{DM}}, m_N, m_{\mathrm{ARP}}), \qquad \text{alert if } s < 0, \qquad \text{severity} = \max(0, -s).$$

During updates, original DM and count references remain active detection checks.

Constant fit features use unit scale. A count-tail breach can decide early. Severity is not an attack probability.

R2 / Device traffic profiles



Device-specific models

Normal protocol proportions differ between devices.

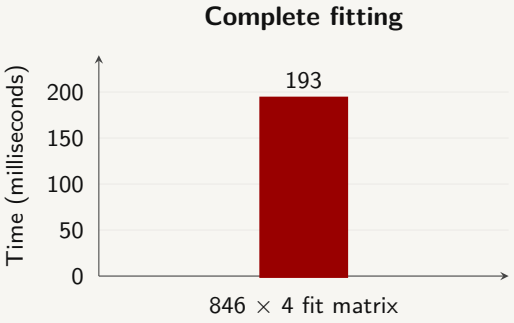
4 exclusive counts
per ten-minute window

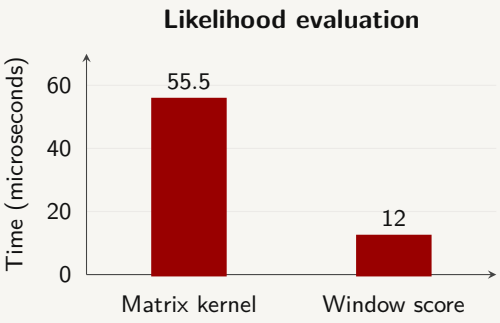
Learn from a trusted benign period; flag unusual later behaviour.

A separate normal model captures each device's traffic pattern.

Fitted means from saved normal-only models. UDP excludes SSDP; bars are model expectations.

R3 / Native Languasco–Migliardi implementation

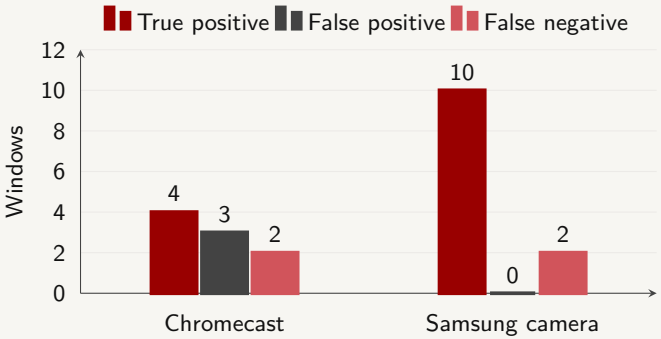




Native integration preserves the sum of independent window likelihoods.

Retained D-Link native-method timings, five batches and warm-up. Fitting excludes input/output. These are recorded measurements.

R4 / Final confusion counts by device



Chromecast
432 windows; 6 attack
TP = 4 FP = 3
FN = 2 TN = 423
FPR = 3/426 = 0.70%

Samsung camera
576 windows; 12 attack
TP = 10 FP = 0
FN = 2 TN = 564
FPR = 0/564 = 0%

Window recall and attack-interval coverage answer different questions.

Static mode, final partitions. A window is attack-labelled if it overlaps an official attack interval.

R5 / Conservative learning guards

Score first
Emit current decision

→

Admit safe evidence
Active + original confidence, volume + ARP guards

→

Check candidate
Enough data, valid fit, bounded movement

$$1/1.25 \leq \alpha_0'/\alpha_{0,t} \leq 1.25 \qquad \text{concentration ratio}$$

$$\max_k |p_k' - p_{k,t}| \leq 0.05 \qquad \text{protocol proportions}$$

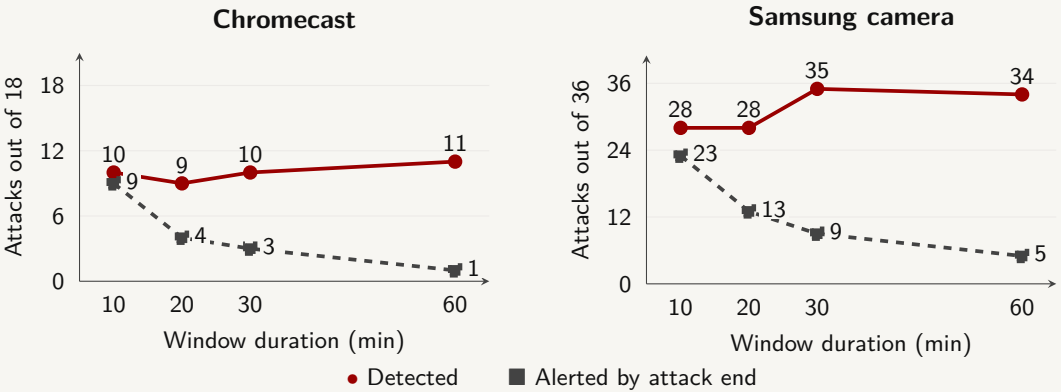
$$|\tau' - \tau_t| \leq 2\,\mathrm{IQR}_0 \qquad \text{threshold movement}$$

Defaults: 288-window buffer; attempt every 144 scored windows; 144 accepted windows required for periodic fitting.

Rejecting a candidate retains the complete active pair; accepted pairs support rollback.

Per-update guards do not establish immunity to stealthy attacks or unlimited cumulative drift. Final replay promoted no candidates.

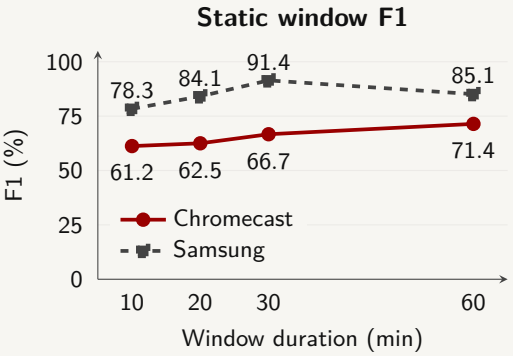
R6 / Window duration: attack coverage

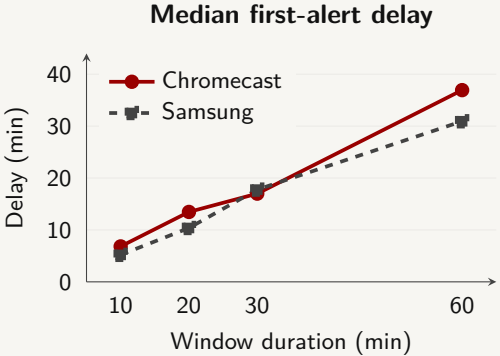


Delivered reference: ten minutes. Longer windows increase response delay.

Static development study; matched traffic within each device; fresh fitting and calibration at every duration.

R7 / Window duration: F1 and detection delay





F1 changes with label granularity; median delay is conditional on detection.

Static development study. Samsung uses May 28–29 to fit and May 30–31 to calibrate at every duration. Lines join measured settings.

R8 / References and project evidence

Statistical learning

Minka, *Estimating a Dirichlet distribution*, 2000 / revised 2012.
Fixed-point estimation of the DM parameters.

Native likelihood method

Languasco and Migliardi, *Computational Statistics* 38, 2023.
doi: 10.1007/s00180-022-01311-7
Al-Haj Baddar, Languasco and Migliardi, *ICNC*, 2025.
doi: 10.1109/ICNC64010.2025.10993688
Al-Haj Baddar, Languasco and Migliardi, *IEEE TPAMI* 47(2), 2025.
doi: 10.1109/TPAMI.2024.3489645

Research and market context

N-BaIoT, Kitsune and DÏoT: neural, ensemble and federated research systems.
Microsoft Defender for IoT and Palo Alto Device Security: representative commercial scope.

Project evidence

Source code, protocols, results and reproducibility records:
`github.com/Pitclair/iot_anomaly_detection`